

Disproof of conjecture 00000000543:
for the three lines $x, y, x + y$ the Orlik–Terao algebra has depth
 ≥ 1 , while $\text{rank} - 1 - \beta = 0$

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2026-10-05

1 The conjecture and the reading

Statement (English, verbatim). Definition: Matroid complexes and their Orlik–Terao algebras. Conjecture: The depth of the OT algebra equals the rank of the matroid minus 1 minus β (a depth-beta invariant formula for hyperplane arrangements). (The Chinese text states the same formula: depth of the OT algebra = rank of the matroid $- 1 - \beta$.)

Reading. For a (central) arrangement $A = \{H_1, \dots, H_m\}$ of distinct hyperplanes in k^n , with matroid $M(A)$ on $E = \{1, \dots, m\}$, the claim is

$$(*) \quad \text{depth OT}(A) = r(E) - 1 - \beta(M(A)).$$

We work over an arbitrary field k of characteristic 0 (e.g. $\mathbb{Q}$). We give an arrangement of three lines in k^2 for which $(*)$ fails with a non-negative right-hand side (Theorem 6), and a second arrangement (four lines) for which the right-hand side is -1 , so that $(*)$ fails for *every* notion of depth with values in $\mathbb{N} \cup \{\infty\}$ (Theorem 7).

2 Definitions (as formalized)

Definition 1 (arrangement, matroid). An arrangement of m hyperplanes in k^n is given by coefficient vectors $v_1, \dots, v_m \in k^n$; the defining linear form of H_i is $\alpha_i = \sum_j v_i(j) x_j \in k[x_1, \dots, x_n]$. The rank function of the matroid is $r(S) = \dim_k \text{span}\{v_i : i \in S\}$ for $S \subseteq E$; the rank of the matroid is $r(E)$. The beta invariant is

$$\beta = (-1)^{r(E)} \sum_{S \subseteq E} (-1)^{|S|} r(S).$$

Definition 2 (Orlik–Terao algebra). $\text{OT}(A)$ is the k -subalgebra of the field of rational functions $k(x_1, \dots, x_n) = \text{Frac } k[x_1, \dots, x_n]$ generated by the reciprocals $y_i = 1/\alpha_i$. It is graded with every y_i in degree one; since it is generated in degree one, its irrelevant (homogeneous maximal) ideal is $\mathfrak{m} = (y_1, \dots, y_m)$.

Definition 3 (depth). A list $(r_1, \dots, r_\ell)$ of elements of a ring R is *R -regular* if each r_j is a nonzero-divisor on $R/(r_1, \dots, r_{j-1})$ and $R/(r_1, \dots, r_\ell) \neq 0$ (Mathlib’s `RingTheory.Sequence.IsRegular`). We set $\text{depth OT}(A) = \sup\{\ell : \text{there is an OT}(A)\text{-regular sequence of length } \ell \text{ in } \mathfrak{m}\} \in \mathbb{N} \cup \{\infty\}$. This is the same pattern as the depth definition in our earlier package for conjecture 00000000537 (file `C537.lean`), which we reuse with credit.

3 The counterexample

Let A be given by $v_1 = (1, 0)$, $v_2 = (0, 1)$, $v_3 = (1, 1)$, i.e. the lines $x = 0$, $y = 0$, $x + y = 0$ in k^2 . Its matroid is the uniform matroid $U_{2,3}$.

Lemma 4 (rank and beta). *If $v_1, \dots, v_m \in k^2$ are nonzero and pairwise linearly independent, then $r(S) = \min(|S|, 2)$ for all S . Hence for A : $r(E) = 2$ and $\beta = \sum_{j=0}^3 \binom{3}{j} (-1)^j \min(j, 2) = 0 - 3 + 6 - 2 = 1$, so the right-hand side of $(*)$ is $2 - 1 - 1 = 0$.*

Proof. $r(S) \leq \dim k^2 = 2$; if $|S| \geq 2$, S contains two independent vectors, so $r(S) \geq 2$; $r(\emptyset) = 0$ and $r(\{i\}) = 1$ since $v_i \neq 0$. The sum depends only on $|S|$. Pairwise independence of $(1, 0)$, $(0, 1)$, $(1, 1)$ is checked coordinatewise. $\square$

Lemma 5 (y_1 is not a unit). *Assume every $c_i := \alpha_i(1, \dots, 1) = \sum_j v_i(j)$ is nonzero (for A : $c = (1, 1, 2)$). Then no y_i is a unit of $\text{OT}(A)$.*

Proof. Let $Q = \prod_i \alpha_i \neq 0$ and $L = k[x][1/Q] \subseteq k(x)$ (Mathlib's `Localization.subalgebra.ofField`). Since $y_i = (Q/\alpha_i) \cdot Q^{-1}$, every $y_i \in L$. The k -algebra map $g : k[x] \rightarrow k[T, T^{-1}]$, $x_j \mapsto T^{-1}$, sends $\alpha_i \mapsto c_i T^{-1}$, hence Q to a unit, so it extends to $\text{ev} : L \rightarrow k[T, T^{-1}]$, and $\text{ev}(y_i) = c_i^{-1} T$ (because $y_i \alpha_i = 1$). By induction over the generators of $\text{OT}(A)$ (constants go to constants; sums and products are preserved), every $f \in \text{OT}(A)$ lies in L and $\text{ev}(f)$ lies in the image of $k[T]$. If $y_i f = 1$ with $f \in \text{OT}(A)$, then $c_i^{-1} T p(T) = 1$ for some $p \in k[T]$; as $k[T] \rightarrow k[T, T^{-1}]$ is injective, $T \cdot c_i^{-1} p = 1$ in $k[T]$, so T would be a unit of $k[T]$, which is false. $\square$

Theorem 6. *Under the hypothesis of Lemma 5, $[y_i]$ is an $\text{OT}(A)$ -regular sequence in $\mathfrak{m}$, so $\text{depth OT}(A) \geq 1$. For the arrangement $\{x, y, x + y\}$ this contradicts $(*)$, whose right-hand side is 0.*

Proof. $\text{OT}(A)$ is a subring of a field, hence a domain, and $y_i \neq 0$ (as $\alpha_i \neq 0$), so y_i is a nonzerodivisor. By Lemma 5, $(y_i) \neq \text{OT}(A)$, i.e. $\text{OT}(A)/(y_i) \neq 0$. And $y_i \in \mathfrak{m}$. So $\text{depth OT}(A) \geq 1 > 0 = r(E) - 1 - \beta$ by Lemma 4; in particular $\text{depth OT}(A)$ is not equal to the integer $r(E) - 1 - \beta$. $\square$

Theorem 7 (a second witness). *For the four lines $x, y, x + y, x + 2y$ in k^2 (matroid $U_{2,4}$): $r(E) = 2$, $\beta = \sum_{j=0}^4 \binom{4}{j} (-1)^j \min(j, 2) = 0 - 4 + 12 - 8 + 2 = 2$, and $r(E) - 1 - \beta = -1$. Hence no $D \in \mathbb{N} \cup \{\infty\}$ equals the right-hand side of $(*)$, whatever reading of “depth” or of “OT algebra” is used.*

Proof. Lemma 4 applies: the four vectors are nonzero and pairwise independent (in characteristic 0). $\square$

Remark 8 (context, not used in the proofs). Proudfoot and Speyer, *A broken circuit ring* (arXiv:math/0410069), study “the ring generated by the inverses of the restrictions of the linear functionals” and state in their abstract: “In particular, $R(L)$ is Cohen-Macaulay”. This is consistent with our finding that the depth of $\text{OT}(A)$ is positive; our proofs above are self-contained and do not rely on it.

4 Lean formalization

File `lean/Conjecture543/Basic.lean` (namespace `C543`, only `import Mathlib`). Definitions: `alpha` (written with the Greek letter in Lean), `y`, `OT` (`Algebra.adjoin k (Set.range (y v))`) inside `FractionRing (MvPolynomial (Fin n) k)`), `gen`, `irr`, `depth`, `rk`, `beta`; witnesses `vA` ($U_{2,3}$) and `vB` ($U_{2,4}$). Main results:

- `not_isUnit_gen`, `isRegular_gen`, `one_le_depth`: Lemma 5 and Theorem 6 for any arrangement with all $c_i \neq 0$.
- `rk_eq_min`, `rk_vA_univ`, `beta_vA`: Lemma 4.
- `witness`: $r(E) = 2 \wedge \beta = 1 \wedge 1 \leq \text{depth} \wedge \neg \exists d \in \mathbb{N}, \text{depth} = d \wedge d = r(E) - 1 - \beta$ for `vA`.
- `conjecture_543_false`:

```
theorem conjecture_543_false :
  NOT forall (m n : Nat) (v : Fin m -> Fin n -> k),
    (forall i, v i != 0) ->
    (forall i j, i != j -> LinearIndependent k ![v i, v j]) ->
    rk v Finset.univ = n ->
    exists d : Nat, depth v = d /\
      (d : Int) = (rk v Finset.univ : Int) - 1 - beta v
```

(ASCII transcription of the Lean statement.) (for every field k of characteristic 0; the hypotheses say: nonzero forms, distinct hyperplanes, essential arrangement).

- `formula_U24`, `not_formula_U24`: Theorem 7.

Axiom audit. `#print axioms for C543.conjecture_543_false, C543.witness and C543.not_formula_U24` reports only `[propext, Classical.choice, Quot.sound]`. No sorry, no native_ decide, no added axioms.